Helping Hand

Preliminary Report: Wearable-Assisted ASL Learning

Brian Paz, Gael Garcia, Kali Schuchhardt, Srinitha Srikanth
Computer Engineering, CEN4908C
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Abstract

Helping Hand is a wearable-assisted American Sign Language (ASL) learning prototype intended to support beginner practice through sensor-based feedback and persistent learner progress. Five resistive flex sensors and a wrist-mounted six-axis inertial measurement unit (IMU) connect to an ESP32-S3, which streams measurements and static predictions over Bluetooth Low Energy (BLE) to a Flutter Android application. The application provides account access, target-specific practice, recording and export tools, and local-first progress storage with Firebase synchronization. This report describes the design, scientific background, preliminary results, engineering constraints, community considerations, and remaining development. The original 36-class static classifier is preserved as a baseline; current firmware packages a separate 34-class model trained from recorded poses. For isolated words, a temporal convolutional network (TCN) was compared with two convolutional/recurrent alternatives using fifteen real recordings of Hello, Please, and Yes. All candidates correctly classified three validation trials; the TCN was selected by the predefined model-size tie-break at 48,324 bytes and subsequently classified three reserved test trials correctly. These small, neutral-orientation results do not establish performance across users or live conditions. Software checks and Android recorded replay support integration, while current physical validation remains incomplete. Two damaged flex sensors have delayed reliable new data collection. Replacement flex sensors remain awaited; the nine-axis IMU and piezo elements have been received for planned orientation sensing and tactile feedback. Near-term work addresses sensor repair, calibration, printed circuit board (PCB) development, recognition robustness, haptic cue mapping, and reproducible end-to-end testing. Educational effectiveness and reliable corrective feedback remain objectives requiring user evaluation.

1 Introduction

Learning ASL requires practice of handshape, movement, orientation, location and nonmanual features. ASL is a natural language with grammar distinct from English; it is not a universal sign system [1]. Helping Hand explores whether a portable sensor glove and companion application can make independent beginner practice more informative between interactions with instructors and fluent signers.

The original design identified hearing beginners as intended learners and proposed tactile correction, modular hardware and structured practice [2]. The prototype combines recognition and saved practice; corrective haptics remain under development. A recent study involving ASL educators identifies structured signing feedback as a useful technology direction while emphasizing pedagogical and cultural considerations [3]. Helping Hand therefore targets a bounded learning aid; recognition of selected signs alone cannot demonstrate ASL competence or improved learning.

2 Background and Technical Basis

A static classifier maps a single sensor vector to a label. Moving signs require temporal structure: the order and duration of observations can distinguish gestures that share individual poses. Lee, Chong and Chung demonstrated word-level ASL recognition using six IMUs and a long short-term memory (LSTM) network [4].

That study supports feasibility, but its fingertip sensing, dataset and evaluation differ from Helping Hand's five flex sensors and single wrist IMU. Published accuracy is not transferable to this prototype.

TCNs use temporal convolutions to combine neighboring observations; dilation extends the temporal context without a recurrent state. Bai, Kolter and Koltun provide a general comparison of convolutional and recurrent sequence models in an author preprint [5]. Helping Hand compares TCN, CNN-GRU and CNN-LSTM on identical recordings and preprocessing. These alternatives combine convolutional neural networks (CNNs) with gated recurrent unit (GRU) or LSTM layers. Trial boundaries must remain intact during splitting: neighboring frames from one sign are correlated and should not appear as independent training and test examples.

Orientation is both a sensing problem and meaningful sign information. Quaternion-based activity-recognition research motivates evaluating frame transformations under sensor rotation [6]; it does not prove that removing orientation helps ASL. Accelerometer/gyroscope fusion can estimate tilt and short-term rotation, while a calibrated magnetometer can provide a heading reference. Kok and Schön demonstrate the importance of magnetic calibration and axis alignment [7]. A magnetic reference may reduce yaw drift, but nearby magnetic disturbances can degrade heading. The nine-axis upgrade will test this tradeoff; no orientation normalization is implemented in the current word demo. Haptic perception also requires testing during movement: Tajdari, Forsyth and Lim found lower vibrotactile identification accuracy during repetitive hand motion [8]. Their results motivate a glove-specific cue study, not a claim that our acquired buzzers provide effective guidance.

3 System Design

3.1 Acquisition, communication and inference

The current glove retains the demo's perfboard-based electronics assembly (Figure 1). It connects five flex channels to ESP32-S3 analog inputs A0–A4 and reads its current accelerometer/gyroscope over I^2C. Firmware targets a 40 Hz stream and records sequence numbers and device timestamps for delivery checks. BLE notifications carry raw flex values, acceleration, angular rate, sensor status and static predictions to one shared Flutter connection service. The Nordic UART-style custom GATT service provides device-to-app notifications and an app-to-device write characteristic [9, 10]. Its text packet contract is project-specific; the write path currently logs commands and does not drive haptics. Sustained packet integrity still requires target-phone measurements.

The static path runs a multilayer perceptron on ESP32. Current firmware defaults to a 34-class recorded model covering digits and letters except motion-based J/Z; the original synthetic 36-class baseline remains separately available. The recorded model uses five calibrated flex inputs, two hidden layers of 128 and 64 units, and a softmax output [11].

The word path runs locally on the phone. Each complete attempt is linearly resampled at 40 Hz using the device clock, center-trimmed or edge-padded to 96 samples (approximately 2.4 seconds), and standardized using training-only statistics. Its eleven features are five raw flex readings followed by three acceleration and three angular-rate channels. A 16-filter projection feeds three residual TCN blocks with two kernel-five causal convolutions per block and dilations 1, 2 and 4. Temporal averaging, a 16-unit dense layer and softmax yield three word scores. Dropout and L2 regularization are already present [12].

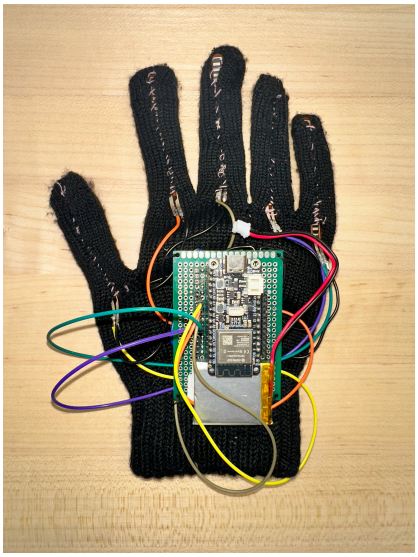


Figure 1: Team prototype photograph used in the October 6 presentation: finger sensor mounting, wiring and glove-mounted perfboard electronics [19]. The photograph documents assembly, not sensor health or integrated haptic feedback.

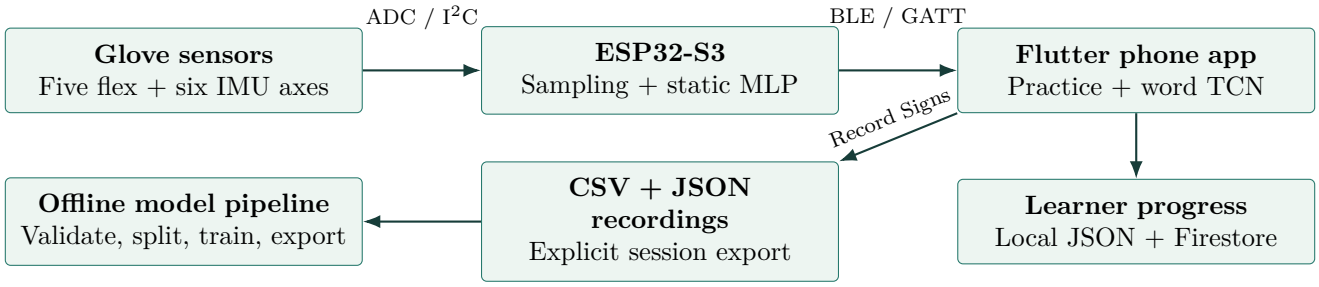


Figure 2: Implemented processing paths. Word inference is on the phone; training is offline. Proposed piezo feedback and the received nine-axis IMU are not shown as integrated components.

3.2 Learner interaction and persistence

Learners sign in, choose Alphabet, Numbers or Words, and open a dedicated practice page. Connection state, predictions, confidence and retry feedback are visible. Static completion requires a matching target with at least 80% confidence for 750 ms and five matching packets, with gaps no greater than 250 ms. Word practice instead uses explicit Start attempt and Finish sign boundaries; completion requires the actual highest-scoring word to match the target at 80% confidence. Invalid packets, interruptions and cancellation discard incomplete attempts [13].

Progress is saved locally under the user's identity, then synchronized to an owner-scoped Firestore document. Firebase handles email/password authentication; raw recordings follow a separate app-private CSV/JSON export workflow [14, 15]. Cloud synchronization uses the phone's Wi-Fi or cellular internet connection; Firebase protects service traffic in transit using HTTPS [16]. The team observed completed lessons in both the app and database during the October demonstrations. Restart persistence, account isolation and offline recovery remain beta validation tasks.

3.3 Planned haptics and risk mitigation

Ten acquired piezo elements are planned: one per finger and five in an X on the back of the hand. They are intended to identify fingers requiring correction and communicate direction. The team confirms the purchased parts match the TIMESETL listing for 27 mm externally driven piezo pickup discs [17]. Driver requirements, pulse patterns, tactile performance and error-to-cue mapping remain unresolved. A word label does not identify which finger should move; a separate correction method must be designed. Initial work will validate one cue and actuator before expanding the array, checking comfort, distinguishability, power and sensor interference.

Damaged thumb and index flex sensors currently block reliable fresh acquisition. Replacement mounting will include contact strain relief, supported by Adafruit's handling guidance [18]. Recalibration and new recordings will address distribution changes. Further risks include confident predictions for unrelated motion, truncated long gestures, background completion of static lessons, and firmware waiting indefinitely for USB serial at startup. Existing records identify these issues; rejection testing, explicit practice lifecycle handling and standalone boot tests are planned mitigations [19].

4 Tools, Constraints and Engineering Standards

PlatformIO with Arduino/C++ builds ESP32 firmware; Flutter/Dart implements the Android interface [20, 21]. TensorFlow/Keras trains the models, with TensorFlow Lite (TFLite) deployment and training-only scaling; Python tests and Flutter unit/widget/integration checks exercise the pipeline [22, 23, 24, 25, 26]. KiCad supports circuit design and board-layout planning, while Git records source revisions [27, 28].

I^2C supplies the sensor bus; current firmware configures 100 kHz, corresponding to Standard-mode's nominal maximum [29]. BLE/GATT defines wireless access to characteristics, with a custom UART-style service layered above it [9, 10]. The received Adafruit ICM-20948 breakout adds an AK09916 magnetometer and board-level regulation/level shifting. It requires different driver code and verification of axes, units, calibration and fusion before providing useful orientation features [30].

Constraints include limited independent recordings, sensor fragility, glove fit, battery capacity, actuator wiring and embedded memory. The recorded static firmware build uses 37.5% of configured RAM and 83.9% of application flash; the word model is deployed on the phone [11]. Listed replacement prices are $7.95 per flex sensor and $19.95 for the IMU as checked October 27; actual expenditure, full bill of materials and battery runtime remain unconfirmed [18, 30].

5 Needs, Impact and Professional Considerations

The initial need remains useful independent practice for hearing beginners, with instructor interaction and authentic ASL examples central to learning. The engineering contribution is integration of wearable sensing, local recognition, recording and account-specific practice, with proposed tactile guidance. A remote Notre Dame ASL-club discussion encouraged beginner progression, calibration and haptic feedback and noted the missing second hand and facial information. The participants were ASL learners reviewing media; their comments do not establish worn comfort, linguistic validation or representative user acceptance [31].

Cultural and social impact. Incorrect automatic feedback could reinforce errors or frustrate learners. Instruction and vocabulary should be reviewed by fluent signers and Deaf educators, including variation in signing and handedness. Deaf-led research identifies biases in sign-language AI agendas and datasets, supporting meaningful community participation [32]. A single glove cannot observe facial grammar, the uninstrumented hand or body-relative hand location reliably. Proposed learning benefits remain unmeasured.

Global impact. The hardware/software pattern may be reusable, but ASL does not represent all sign languages. Other languages require their own linguistic review, vocabulary, recordings and evaluation [1].

Economic and environmental impact. Requiring a glove and compatible phone may limit access. Replacement cost and downtime are practical constraints, and low purchase prices alone do not establish affordability. Replaceable sensors could extend service life; failed sensors, wiring and batteries still create waste. Repairability and appropriate disposal are relevant to the wider electronics-waste problem, but no life-cycle benefit has been measured [33].

Ethical and professional practice. Collection protocols call for consent, pseudonymous signer identifiers and controlled export of recordings. Account credentials belong to Firebase Authentication rather than sensor files. Actual sharing permissions and account isolation need review/testing. Reporting tiny datasets honestly, preserving failures, checking wearable electrical and mechanical limits, and crediting sources follow professional obligations to privacy, public welfare and realistic technical claims [34]. Future work should evaluate task usability before making educational-effectiveness claims; improved recognition alone does not demonstrate learning or retention.

6 Results and Evidence Limits

6.1 Alpha software and recording evidence

At commit `cce9d38` on September 10, project records report successful Flutter analysis, 16 Flutter tests, Android debug compilation, ESP32-S3 compilation and 23 backend tests [19]. These checks exercised software correctness and buildability. Historical videos show physical flex telemetry reaching an Android phone and changing its visualization, but do not provide a controlled accuracy study. Missing location/device metadata limit exact reproduction.

The September 17 static pilot saved 35 trials and 5,518 valid frames from one signer at neutral orientation. Record Signs retained raw packets, labels, device timestamps and session metadata; export review checked validity, ordering and sequence gaps. Documented rates were approximately 39.7–40.5 Hz, with no observed sequence gaps [35]. This supports recording/export feasibility under that session, not model accuracy; most labels have one trial.

6.2 Recorded static and word evaluation

The October 6 recorded static model used 52 whole trials, with seed 42 assigning 44 to training and four each to validation/test. Only A, B, 1 and 3 had held-out trials; the other 30 classes remained training-only. Weighted training used Adam, batch size 32, an 80-epoch cap and validation-loss patience 8. The restored checkpoint correctly classified all 615 test frames and four trial-averaged predictions. Retraining reused the split during calibration revision, making this a regression check, not fresh untouched-test or all-class evidence. Calibration also reduced overall training fit and left errors including C/X, E/A and W/6 confusions [11]. This motivates fresh post-repair collection rather than relying on priority-sign success. Reproduction uses the two retained static exports, replaces older A/B/1/3 trials with the newer session, applies fourfold priority class weighting, and fits the scaler on training groups only. The recorded environment was Windows CPU with TensorFlow 2.21.0 and Keras 3.14.0; calibration, source hashes and frozen trial IDs are retained in the versioned record.

On October 5, the word comparison used fifteen trials, five per word, with seed 7 assigning three training, one validation and one test trial per word. All candidates used identical train-only scaling, 96-sample windows, 16-filter/ unit configurations, dropout 0.25, L2 0.0001, Adam learning rate 0.001, batch size 3, a 150-epoch cap and patience 20. Selection ranked validation accuracy, macro F1, file size and desktop latency, in that order. Only the selected candidate was then evaluated on reserved tests [12].

Candidate	Validation	Macro F1	TFLite bytes	Median ms
TCN	3/3	1.00	48,324	0.0488
CNN-GRU	3/3	1.00	180,212	0.0442
CNN-LSTM	3/3	1.00	159,320	0.0611

Table 1: Same-split word comparison on three validation trials. Desktop CPU timings exclude capture, BLE, preprocessing and UI; they are not phone latency.

The expected outcome was a compatible, compact candidate. TCN won the size tie-break and correctly classified all three reserved tests, yielding macro F1 1.00 and one correct trial per class in the confusion matrix. Top-five accuracy is inapplicable to three classes. These neutral recordings came from the reported demo wearer; metadata signer IDs do not establish independent users. Desktop timing used one TFLite thread, ten warmups and 100 measured copy/invoke/output-copy calls. The input is the retained word CSV/manifest pair; reserved trials are `word_hello_0008`, `word_please_0011` and `word_yes_0005`. Evaluation compares the highest-scoring label with each trial's label, then computes class F1 and the confusion matrix. Device timestamps supply resampling because repeated phone receive times were flagged. Full generated runs require the team's retained artifact backup [12].

6.3 Team-reported demonstration and beta integration

The demonstrations checked recognition, lesson completion and database storage. During October 5–6 demonstrations in the NSC senior design lab, the team used the documented Fairphone Gen 6/Android 15 configuration. Reported successes included letter practice for A, B, C, L, S, U and Y; all digits were attempted. Hello and Please were recognized and their lessons marked complete. Gael verified saved progress by observing green completion indicators for practiced letters, numbers and words in the app and checking corresponding records in the Firebase database. These expected outcomes were observed; restart persistence and account isolation remain unverified. Faulty thumb and index sensors produced bad readings and made some signs difficult. Per-sign trial counts and exact software revisions remain unrecorded [36]. These are team-reported application observations, not controlled accuracy or measured learning.

October software integration checks covered Python/Dart input parity, completion/cancellation, progress and widgets: 42 focused checks passed. Three native recorded replays passed on a Pixel 8 API 36 emulator using the packaged TCN, with predictions within 0.02 percentage points of Python; the ordinary Android debug APK also built [13]. Replay formed complete attempts from the already evaluated recordings, ran the packaged interpreter and compared its output with the Python reference. The expected matching predictions were obtained without a physical glove or Firebase sign-in. It verifies deployment consistency, not additional held-out accuracy.

Post-repair recognition, sustained BLE delivery, intended-phone latency, offline/restart recovery and external-user task tests remain **Not run**. Two broken flex sensors explain the current collection block. The T1/T2 procedures require recording build, device/OS, firmware, hardware, date, tester, expected result, observed measurements and evidence for future runs [37].

7 Current Status and Completion Plan

Account screens, practice routes, progress storage, recording/export, the static classifier and three-word TCN integration are implemented. Current hardware revalidation and instructional-media review remain incomplete. Since the October 6 demonstration, circuit design and board-layout planning have continued in KiCad; the October 22 schematic (Figure 3) documents progress, while saved board files contain no placed/routed lay-

out. The current TCN review prioritizes broader data, complete-gesture windows, unrelated-motion rejection and sensor/orientation robustness. It establishes improvement directions, not measured gains [19, 38].

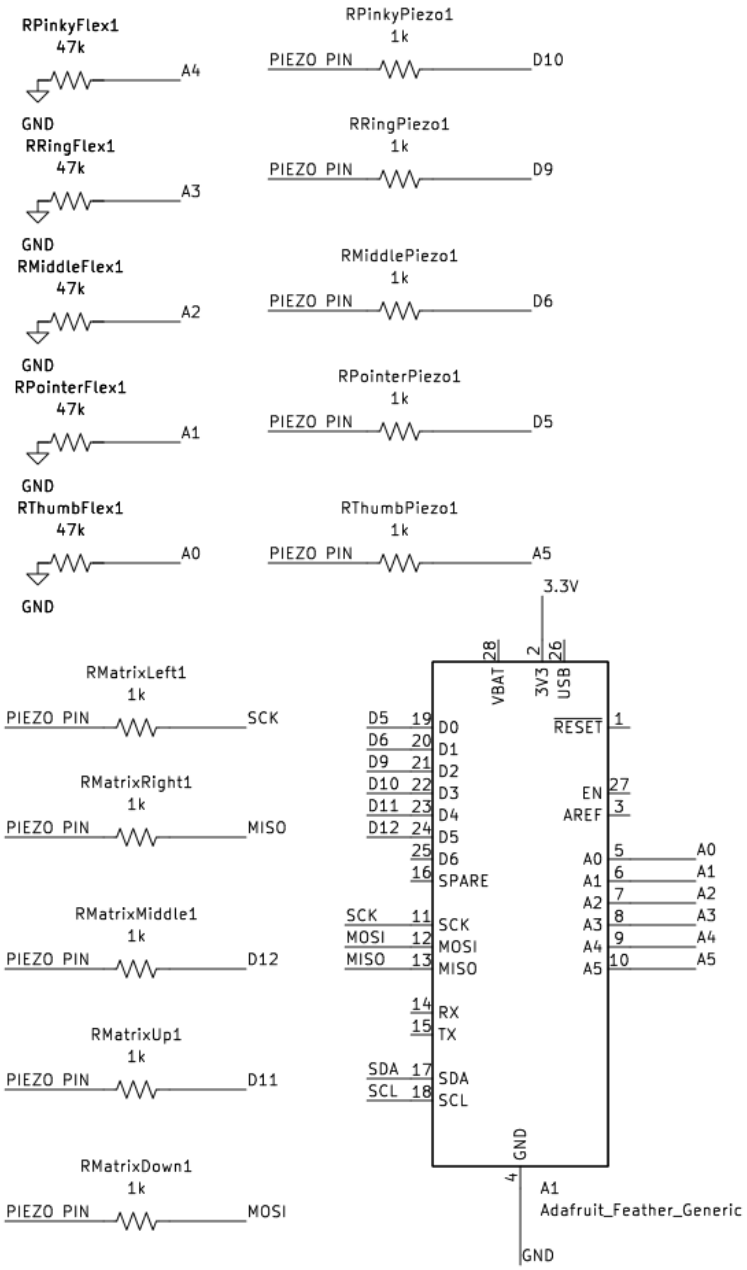


Figure 3: October 22 KiCad connection draft, exported from the team's saved schematic [19]. It shows flex-resistor branches and tentative ten-piezo connections, not the complete as-built circuit. The generic Feather symbol, board-specific pins, repeated piezo net labels and missing sensor/driver circuits require review before layout.

Date/window	State	Milestone / dependency
September 10–17	Recorded	Alpha builds/tests; real static recording/export pilot.
October 2–6	Demonstrated	Account/practice implementation, recorded model training and phone integration; sign completion and database progress observed. Controlled live validation incomplete.
October 22–28	In progress	KiCad schematic/layout planning and TCN review; glove remains on perfboard. Piezo elements and ICM-20948 received; flex repair awaited.
Next two weeks	Planned	Replace/calibrate flex sensors; integrate ICM-20948; complete circuit/driver design and initial haptic cues. Sequence depends on delivery and bench checks.
Before November 19	Proposed target	Collect fresh trials, evaluate revised features/model, verify live practice and progress, and test cue perception.
November 19	Scheduled	Post-mortem presentation.
December 3	Tentative	Evening CpE Showcase, approximately 6 p.m., WERT Lobby.

Table 2: Completed work and team-reported roadmap as of October 28. Future engineering dates are targets.

Repair will establish per-finger ranges before new data collection. IMU work will verify all nine channels and compare calibrated fusion against a baseline, preserving meaningful palm orientation. New input features require matching training and phone preprocessing. Haptic work must connect a defined correction signal to bounded actuator commands and validate perception with sensors monitored both during and after activation. Final checks should include wrong signs, idle motion, disconnects, restart, account isolation and repeated latency measurements. Independent trials and held-out users are needed before broader recognition claims. Piezo drive specifications, mounting and tactile suitability still need verification.

8 Conclusion

Helping Hand integrates wearable sensing, saved practice, recording tools and a compact three-word TCN. Software checks, offline experiments and team-reported demonstrations support the prototype; live reliability, generalization and learning benefits require further evaluation. Despite sensor damage, circuit planning and model review have continued. The next phase will calibrate the repaired glove, integrate the received ICM-20948, evaluate recognition changes and prototype directional piezo cues. Reproducible system/user tests and ASL community expertise should guide future vocabulary and feedback development.

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